

My First Deep-Learning Project — One-Pager

Name: _____ Assay / instrument: _____

Fill one page for a real project in *your* lab. The goal is a plan you could hand a collaborator — and an honest answer to “do I even need deep learning?” Keep it concrete: one example, real shapes and units, a split with no leakage, and clinical-grade metrics (Lecture 10).

1 • Problem

What decision or measurement would the model make, and who uses the output? E.g. “flag likely-resistant isolates from the MALDI spectrum before culture comes back.”

2 • Data & representation

Describe one example exactly: shape, units, how many samples, how they are labeled, and how imbalanced. E.g. “one 6000-dim TIC-normalized spectrum; ~700 samples; 5–25% positive; labeled by AST.”

3 • Architecture

Which model, and why does it fit this representation? E.g. “1D-CNN — peaks are local patterns anywhere along m/z, like Lab 2 / Track A”; or MLP for a tabular panel; or transfer/re-head an existing model (Track C).

4 • Validation plan

What split (no leakage — split by patient/batch, not by row!), which metrics, and what target? E.g. “stratified 80/20 by patient; sensitivity / specificity / AUROC + confusion matrix; ≥ 0.90 sensitivity at a fixed specificity, threshold chosen on a validation set — never a single accuracy number.”

5 • Why not a simpler model?

What baseline must deep learning beat to be worth it — logistic regression, a single-peak threshold, or always-guess-the-majority-class? If a one-line rule already hits your target, deep learning may be the wrong tool — and saying so is a valid, honest conclusion (the Lecture 10 mindset).

Discussion D5: share this page with a neighbor. Pressure-test each other on leakage (could val info leak into training?), the honest majority-class baseline, and the clinical cost of a false negative vs. a false alarm — does your threshold reflect it?